David R. Miller

PhD Candidate - University of British Columbia

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Education

University of British Columbia *Vancouver, BC, Canada*
PhD, Astronomy *2022–Present*
Supervisors: Dr. Jeremy Heyl and Dr. Harvey Richer (1944–2023)
Thesis: Probing Stellar Evolution With White Dwarfs in Open Clusters
Expected Completion: Spring 2027

University of British Columbia *Vancouver, BC, Canada*
MSc, Astronomy *2020–2022*
Supervisor: Dr. Harvey Richer (1944–2023)
Thesis: The White Dwarf Initial-Final Mass Relation: Ultramassive White Dwarfs Ejected from Young Open Star Clusters

Simon Fraser University *Burnaby, BC, Canada*
BSc, Physics Honours (with distinction) *2016–2020*
Supervisor: Dr. Levon Pogosian
Thesis: Dark Sector Cosmic Fluid Interactions as a Solution to the Observed Disagreement in Measurements of the Hubble Expansion Rate

Awards

UBC Four Year Fellowship (4YF), $98,000 *2023–2027*
President's Academic Excellence Initiative PhD Award, $4,800 *2022–2026*
Dunlap Institute Summer Undergraduate Research Program Fellowship (SURP), $9,500 *2019*
NSERC Undergraduate Student Research Award (USRA), $5,250 *2018*
Ken Caple College Transfer Entrance Scholarship, $3,500 *2016*

Observing Program Leadership

Led and authored 15 successful competitive observing proposals as PI, with one additional jointly developed program as Co-PI, across CFHT, Gemini, and HST.

Recent Highlights

Gemini	*Escaped white dwarfs as probes of the high-mass white dwarf initial-final mass relation*	*2026*
HST (Co-PI)	*Hidden in Plain Sight? Unraveling the Potential Merger Histories of Ultra-Massive White Dwarfs in the Initial-Final Mass Relation*	*2026*
Gemini	*Tracing contamination and stellar interactions in the white dwarf populations of open clusters and their tidal tails*	*2026*
CFHT	*Probing the mass gap in the white dwarf initial-final mass relation with the rich intermediate-aged open cluster NGC 2360*	*2026*
Gemini	*Probing Accelerated Cooling and Formation in a White Dwarf Binary from a Young Open Cluster*	*2026*

Publications

First Author (4)

1. **Miller, D. R.**, Heyl, J., and Tremblay, P-E. Investigating the Observational Progenitor Mass Gap in the White Dwarf Initial-Final Mass Relation. I. Cluster Census and Characterization of the First White Dwarfs in the Gap. 2026, *ApJ*, 1007:183.
2. **Miller, D. R.**, Caiazzo, I., Heyl, J., et al. The White Dwarf Initial-Final Mass Relation from Open Clusters in Gaia DR3. 2026, *ApJ*, 996:69.
3. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. An Extremely Massive White Dwarf Escaped From the Hyades Star Cluster. 2023, *ApJL*, 956:L41. **Featured in:** AAS Nova, Space.com, Universe Today.
4. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. The Ultramassive White Dwarfs of the Alpha Persei Cluster. 2022, *ApJL*, 926:L24.

Second/Third Author (3)

1. Jadhav, V. V., Kroupa, P., **Miller, D. R.**, Sahu, S., and Franktisek, D. The white dwarf population of open clusters and their tidal tails: Tracers of contamination and stellar interactions. 2026, *A&A*, accepted.
2. Yan, H., **Miller, D. R.**, Zhao, J., Guo, J., and Li, C. Escaped White Dwarf Candidates from Open Clusters. 2026, *ApJS*, 286:5.
3. Yan, H., Wang, L., **Miller, D. R.**, et al. Searching for White Dwarf Candidates Formed Through Binary Evolution in Star Clusters. 2026, *ApJ*, 999:167.

Contributing Author (4)

1. Cristea, A. A., Caiazzo, I., Cunningham, T., Raymond, J. C., Vennes, S., Kawka, A., Desai, A., **Miller, D. R.**, et al. A half ring of ionized circumstellar material trapped in the magnetosphere of a white dwarf merger remnant: A new class of white dwarf merger remnants with X-ray emission. 2026, *A&A*, 706, A188.
2. Côté, P., et al. (including **Miller, D. R.**). The CASTOR mission. 2025, *JATIS*, 11(4).
3. Richer, H. B., Caiazzo, I., Du, H., Grondin, S., Hegarty, J., Heyl, J., Kerr, R., **Miller, D. R.**, and Thiele, S. Massive White Dwarfs in Young Star Clusters. 2021, *ApJ*, 912:165. *(Author order alphabetical after the first two)*
4. Tzanavaris, P., Gallagher, S. C., Ali, S., **Miller, D. R.**, Pentinga, S., and Johnson, K. E. Cosmic Pathways for Compact Groups in the Milli-Millennium Simulation. 2019, *ApJ*, 871:242.

Manuscripts in Preparation (4)

1. **Miller, D. R.**, Yan, H., et al. Characterization of Candidate High-Mass White Dwarf Escapees Candidates from Open Clusters. In prep.
2. Yan, H., **Miller, D. R.**, Li, C., et al. Photometric and Spectroscopic Analysis of an Extremely Low-Mass White Dwarf Binary. In prep.
3. **Miller, D. R.**, Caiazzo, I., and Heyl, J. Atypical White Dwarfs in Open Clusters. In prep.
4. **Miller, D. R.**, Heyl, J. A Spectroscopic Census of Cluster White Dwarfs from DESI, LAMOST, and SDSS. In prep.

Select Talks and Presentations

Klosterneuburg, Austria	**24th European White Dwarf Workshop**	*Contributed Talk*	*2026*
Halifax, Canada	**CASCA 2025**	*Contributed Talk*	*2025*
Burnaby, Canada	**Cosmology Seminar,** Simon Fraser University	*Invited Talk*	*2025*
Barcelona, Spain	**23rd European White Dwarf Workshop**	*Contributed Talk*	*2024*
Padova, Italy	**EAS 2024**	*e-Poster*	*2024*
Toronto, Canada	**CASCA 2024: Harvey Richer Special Memorial Session**	*Invited Talk*	*2024*
Vancouver, Canada	**Astronomy Seminar,** University of British Columbia	*Invited Talk*	*2022*
Tübingen, Germany	**22nd European White Dwarf Workshop**	*Contributed Talk*	*2022*

Teaching

Note: After 2023, I did not take on additional TA assignments because my graduate fellowship removed the requirement and my supervisor wanted me to prioritize research. In 2026, I was specifically asked to TA the biennial graduate Stellar Astronomy course, reflecting my standing as the department's leading graduate student in stellar astrophysics.

Astro 508: Stellar Astronomy, TA — *2026*
Sole TA for this graduate course. Duties include holding office hours, supporting student learning, and managing grading.

Astro 310: The Solar System, Head Tutorial TA — *2022–2023*
Oversaw tutorials, managed the course gradebook, held office hours, and revised tutorial materials now used in subsequent offerings.

Astro 311: Stars and Galaxies, Tutorial TA — *2023*

Astro 102: Intro to Stars and Galaxies, Tutorial TA — *2022*

Phys 131: Energy and Waves, Tutorial TA — *2021*

Astro 101: Introduction to the Solar System, Tutorial TA — *2020–2021*
Ran and supported tutorials, held office hours, and marked coursework.

Academic Service & Leadership

JOURNAL REFEREE – *ApJ* — *2026–present*

VOLUNTEER – Testing Gravity, *Vancouver, BC* — *2019, 2025*

SOLE ORGANIZER – Compact Object Journal Club, *University of British Columbia* — *2021–2024*

PROBLEM SETTER AND JUDGE – High-Energy Astrophysics Olympiad, *University of British Columbia* — *2022*

Outreach

PUBLIC TALKS

Space Camp Saturdays, *H. R. MacMillan Space Centre, Vancouver, BC* — *2026*

Science Rendezvous, *University of British Columbia* — *2026*

Public Engagement

Astronomy on Tap, *University of British Columbia* — *2024–2025*

Astronomy on Tap, *University of Toronto* — *2019*

Cronyn Observatory, *University of Western Ontario* — *2018*

Research Affiliations & Professional Development

Legacy Survey of Space and Time (LSST), *Canada* — *2025–present*

- Provisional international data rights holder through Canada's in-kind contribution to the Rubin Observatory.

Summer School in Statistics for Astronomers, *Penn State University* — *2026*

- Training in statistical inference, regression and model selection, resampling, Bayesian analysis, MCMC and nested sampling, time series analysis, spatial statistics, and machine learning in R.

ML4Astro Workshop – ICML 2025, *Vancouver, BC* — *2025*

- Focused on modern machine-learning methods in astrophysics, including foundation models, simulation-based inference, anomaly detection, and AI-assisted scientific workflows.

Testing Gravity Pre-Conference School, *Vancouver, BC* — *2025*

- Advanced review of gravitation and cosmology in preparation for the Testing Gravity conference.

CRAQ Summer School on *Space Astronomy*, *Montreal, QC* — *2023*

- Training in space-based astronomy, satellite operations, and the development of astronomical missions and instrumentation.

CASTOR Science Team - Stellar Astrophysics Working Group, *Canada* — *2022–2023*

- Contributed to development of the stellar astrophysics science case for CASTOR, a proposed Canadian-led UV-optical space telescope.

Testing Gravity Pre-Conference School, *Vancouver, BC* — *2019*

- Introductory training in gravitation and cosmology in preparation for the Testing Gravity conference.

CRAQ Summer School on *Large-Scale Astrophysics*, *Montreal, QC* — *2018*

- Training in galaxy formation and evolution, galaxy clusters, and the use of large-scale structure as a cosmological probe.